

Sessions:-6

Q5. Explain in your own words the difference between encapsulation and abstraction, using examples from any social media app you use daily (like Instagram or WhatsApp).

Encapsulation and **abstraction** are both Object-Oriented Programming (OOP) concepts, but they have different purposes.

Encapsulation

Encapsulation means **keeping data private and allowing access to it only through controlled methods** (getters, setters, or other functions). It protects data from being changed incorrectly.

Example (WhatsApp):

Your phone number, profile picture, and account settings are stored securely. You cannot directly change another user's data. Instead, WhatsApp provides options like **Edit Profile** or **Change Privacy Settings**, which safely update your information.

Abstraction

Abstraction means **hiding the complex internal workings and showing only the necessary features** to the user.

Example (Instagram):

When you tap the **Like** button on a post, Instagram simply shows that the post has been liked. You don't see the complex processes happening behind the scenes, such as sending the request to the server, updating the database, and notifying the post owner. Those implementation details are hidden from you.